

- 7.1.4.** a. Find all functions from $X = \{a, b\}$ to $Y = \{u, v\}$.
 b. Find all functions from $X = \{a, b, c\}$ to $Y = \{u\}$.
 c. Find all functions from $X = \{a, b, c\}$ to $Y = \{u, v\}$.

- a. Find all functions from $X = \{a, b\}$ to $Y = \{u, v\}$.

Let $f : X \rightarrow Y$

Domain of $f = \{a, b\}$

Co-domain of $f = \{u, v\}$

Definition of $f : X \rightarrow Y$:

- (1) Every element in X is related to some element in Y
- (2) No element in X is related to more than one element in Y

Hence:

$$f_1(a) = u$$

$$f_1(b) = v$$

Range of $f_1 = \{u, v\}$

$$f_2(a) = v$$

$$f_2(b) = u$$

Range of $f_2 = \{v, u\}$

$$f_3(a) = u$$

$$f_3(b) = u$$

Range of $f_3 = \{u\}$

$$f_4(a) = v$$

$$f_4(b) = v$$

Range of $f_4 = \{v\}$

Solution: There are 4 functions from $X = \{a, b\}$ to $Y = \{u, v\}$.

- b. Find all functions from $X = \{a, b, c\}$ to $Y = \{u\}$.

Let $f : X \rightarrow Y$

Domain of $f = \{a, b, c\}$

Co-domain of $f = \{u\}$

Definition of $f : X \rightarrow Y$:

- (1) Every element in X is related to some element in Y
- (2) No element in X is related to more than one element in Y

Hence:

$$f(a) = u$$

$$f(b) = u$$

$$f(c) = u$$

$$\text{Range of } f = \{u\}$$

Solution: There is one function from $X = \{a, b, c\}$ to $Y = \{u\}$

c. Find all functions from $X = \{a, b, c\}$ to $Y = \{u, v\}$.

Let $f : X \rightarrow Y$

Domain of $f = \{a, b, c\}$

Co-domain of $f = \{u, v\}$

Definition of $f : X \rightarrow Y$:

- (1) Every element in X is related to some element in Y
- (2) No element in X is related to more than one element in Y

Hence:

$$f_1(a) = u$$

$$f_1(b) = u$$

$$f_1(c) = u$$

$$\text{Range of } f_1 = \{u\}$$

$$f_3(a) = u$$

$$f_3(b) = v$$

$$f_3(c) = v$$

$$\text{Range of } f_3 = \{u, v\}$$

$$f_5(a) = v$$

$$f_5(b) = u$$

$$f_5(c) = v$$

$$\text{Range of } f_5 = \{u, v\}$$

$$f_7(a) = v$$

$$f_7(b) = v$$

$$f_7(c) = u$$

$$\text{Range of } f_7 = \{u, v\}$$

$$f_2(a) = v$$

$$f_2(b) = v$$

$$f_2(c) = v$$

$$\text{Range of } f_2 = \{v\}$$

$$f_4(a) = v$$

$$f_4(b) = u$$

$$f_4(c) = u$$

$$\text{Range of } f_4 = \{u, v\}$$

$$f_6(a) = u$$

$$f_6(b) = v$$

$$f_6(c) = u$$

$$\text{Range of } f_6 = \{u, v\}$$

$$f_8(a) = u$$

$$f_8(b) = u$$

$$f_8(c) = v$$

$$\text{Range of } f_8 = \{u, v\}$$

Solution: There are 8 functions from $X = \{a, b, c\}$ to $Y = \{u, v\}$